Supplementary Materials for " Hierarchical organization of the immunoglobulin M receptor interface across distinct oligomeric states"

Table of Contents

Item	Description
Table S1	Global MD metrics (RMSD, Rg, SASA, Hbonds) with bootstrap 95% CI for all systems
Table S2	MSM validation key parameters (lag time, ITS, CK error, eigenvalue gap)
Table S3	Macrostate populations (%) for WT, D111A, T110A/D111A, and T60A/S63A
Table S4	Mean first passage time (MFPT) matrices (ns) for four systems
Figures S1-S5	High-resolution supplementary figures supporting the MD simulation and MSM analyses
Figures S6	Overview of the integrative computational workflow

Description of Supplementary Files

Tables S1–S4 (supplementary_tables/ folder):

Table S1 (TableS1_global_MD_metrics.xlsx). Backbone RMSD, radius of gyration, solvent-accessible surface area, and number of hydrogen bonds computed over the equilibrated period (20–200 ns) for each system. Values are reported as mean ± 95% bootstrap confidence interval.

Table S2 (TableS2_MSM_validation.xlsx). The chosen lag time (1 ns), first five implied timescales, average Chapman–Kolmogorov test error, and eigenvalue gap after the fifth eigenvalue. These metrics support the fivemacrostate coarse-graining.

Table S3 (TableS3_Macrostate_Populations.xlsx). Equilibrium populations (%) of the five metastable macrostates (M1–M5) for all systems, obtained by PCCA+ coarse-graining of the pooled transition matrix. Corresponds to Fig. 9A.

Table S4 (TableS4_MFPT.xlsx). Complete MFPT matrices (ns) between all five macrostates for each system. Values exceeding 10^6 ns indicate kinetically inaccessible transitions on the simulation timescale. Corresponds to Fig. 9C.

Figures S1–S5 (supplementary_figures/ folder):

Fig. S1 (FigS1_RMSF_both_chains.jpg). Cα RMSF predicted from 200 ns simulations for a single FcμR-D1 chain (chain C) and its binding IgM-Fcμ Cμ4 domain (chain B). A dashed vertical line separates the two chains. The simulations suggest that mutation-dependent increases in flexibility are predominantly localized around Thr60, Ser63, Thr110, and Asp111.

Fig. S2 (FigS2_Hbond_salt_survival.jpg). Survival probability of interfacial hydrogen bonds (left) and salt bridges (right) as a function of lag time. The characteristic lifetime τ is shown. WT and D111A are predicted to exhibit the slowest decay, while T110A/D111A and T60A/S63A are predicted to display accelerated loss.

Fig. S3 (FigS3_DSSP_heatmap.jpg). DSSP secondary structure profiles over 200 ns. Colors code for helix, strand, turn, and coil. The calculations indicate that all systems maintain a stable secondary structure.

Fig. S4 (FigS4_PCA_FEL.jpg). PCA-based free energy landscapes for each system, displayed with plasma colormap and white contour lines as in Fig. 8B. A single dominant basin is predicted for all cases; T60A/S63A is predicted to display the widest basin.

Fig. S5 (FigS5_MSM_validation.jpg). MSM validation. **(A)** Implied timescales are predicted to converge near 1 ns. **(B)** The Chapman–Kolmogorov test suggests good agreement between predicted and estimated probabilities. **(C)** An eigenvalue gap after the fifth eigenvalue is predicted. **(D)** Timescale separation is predicted to support five macrostates.

Fig. S6. Integrative computational framework for dissecting FcμR–IgM recognition. Starting from seven experimentally resolved FcμR–IgM complex structures, the study integrates structural comparison, interface contact analysis, evolutionary conservation assessment, computational alanine scanning, molecular dynamics simulations, and Markov state modeling. These complementary analyses evaluate sequence conservation, energetic contributions, and conformational dynamics of the FcμR–IgM interface, providing a multiscale framework for investigating receptor recognition across distinct IgM oligomeric states. Detailed procedures for each analysis are described in Materials and Methods.